

MA 548 : Abstract Algebra

Course Code: MA 548

Course Name: Abstract Algebra

L-T-P-C: 3-1-0-4

Students intended for: M.Sc./M.S./PhD/B.Tech

Elective or Core: Discipline Elective for M.Sc. Applied Mathematics and Elective for Other Disciplines

Prerequisites:

Mutual Exclusion: MA 251

Approval: 59th BoA

Course Contents

- **Unit I:** Binary operation, and its properties, Definition of a group, Examples and basic properties, Subgroups, Cyclic groups, Order of a group. [6 Lectures]
- **Unit II:** Permutation groups, Isomorphism and its properties, Cayley's theorems, Coset of a subgroup, Lagrange's theorem. [6 Lectures]
- **Unit III:** Normal subgroups, Quotient group. Homomorphisms, Kernel Image of a homomorphism, Isomorphism theorems, Direct product of groups, Structure of finite abelian groups. [6 Lectures]
- **Unit IV:** Group action and class equation, Sylow's Theorems and its applications, Solvable and Nilpotent groups, Lower and Upper central series. [6 Lectures]
- **Unit V:** Rings: definition, examples and basic properties, Zero divisors, Integral domains, Fields, Characteristic of a ring, Quotient field of an integral domain, Subrings, Ideals, Quotient rings. [6 Lectures]
- **Unit VI:** Isomorphism theorems, Prime ideal, Maximal ideals, Ring of polynomials, Division Algorithm and its Consequences, Irreducibility test, Prime elements, Irreducible elements and their properties, UFD, PID and Euclidean domains. [8 Lectures]
- **Unit VII:** The Fundamental Theorem of Field Theory, Splitting Fields, Zeros of Irreducible Polynomials. [4 Lectures]

Laboratory/practical/tutorial Modules:

Weekly tutorials are required to practice the questions based on the contents discussed in the class

Text books:

1. Joseph Gallian, **Contemporary Abstract Algebra**, 7th Edition.
2. N. Herstein, **Topics in Algebra**, Wiley Eastern Ltd., 1975.

References:

1. M. Artin, **Algebra**, Prentice Hall of India, 1994.



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2. J. B. Fraleigh, **A first Course in Abstract Algebra**, Narosa Publishing House, 2003.
 3. Klima, Sigmon and Stitzinger, **Applications of Abstract Algebra with Maple and MATLAB**, Second Edition, 2006.